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प्रश्न-पत्र कोड
Q.P. Code

55(B)

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

भौतिक विज्ञान (सैद्धान्तिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

PHYSICS (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

नोट	NOTE
(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 31 हैं।	(I) Please check that this question paper contains 31 printed pages.
(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।	(II) Please check that this question paper contains 33 questions.
(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।	(III) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।	(IV) Please write down the serial number of the question in the answer-book before attempting it.
(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।	(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions carefully and follow them :

- (i) *This question paper contains **33** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** sections – **Sections A, B, C, D** and **E**.*
- (iii) *In **Section A** – Questions no. **1** to **16** are Multiple Choice type questions. Each question carries **1** mark.*
- (iv) *In **Section B** – Questions no. **17** to **21** are Very Short Answer type questions. Each question carries **2** marks.*
- (v) *In **Section C** – Questions no. **22** to **28** are Short Answer type questions. Each question carries **3** marks.*
- (vi) *In **Section D** – Questions no. **29** and **30** are case study based questions. Each question carries **4** marks.*
- (vii) *In **Section E** – Questions no. **31** to **33** are Long Answer type questions. Each question carries **5** marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.*
- (ix) *Use of calculators is **not** allowed.*

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

SECTION A

1. When a glass rod is rubbed with a silk cloth, it acquires a positive charge of 4.8×10^{-17} C. In the process, the number of electrons and protons transferred are respectively :
(A) 0, 300 (B) 0, 225
(C) 150, 150 (D) 300, 0
2. The emf and internal resistance of a cell are 3 V and 0.2Ω respectively. It is connected to an external resistor of 5.8Ω . The potential difference across the cell will be :
(A) 1.2 V (B) 2.0 V
(C) 2.9 V (D) 3.2 V
3. The magnetic field due to a current carrying circular coil at a point far off on its axis at a distance 'r' from the centre of the coil is proportional to :
(A) $\frac{1}{r}$ (B) $\frac{1}{r^2}$
(C) $\frac{1}{r^{3/2}}$ (D) $\frac{1}{r^3}$
4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :
(A) $\mu_0 NI$ (B) $\frac{\mu_0 NI}{L}$
(C) $\frac{\mu_0 NI}{4\pi L}$ (D) $\mu_0 NI$

5. A circular loop of radius 'a' carries a current $3I$ through it. Another concentric circular loop of radius $2a$ carrying a current I is placed perpendicular to the first loop. The dipole moment of the equivalent magnetic dipole for the system is :
- (A) πIa^2 (B) $5Ia$
 (C) $5\pi Ia^2$ (D) $7\pi Ia^2$
6. The magnetic flux ϕ (in Wb) through a closed loop of resistance 3.0Ω varies with time t (s) as $\phi = 6t^2 + 3.0t + 5$. The value of the induced current in the loop at $t = 1.0$ s is :
- (A) 15.0 A (B) 3.0 A
 (C) 4.0 A (D) 5.0 A
7. The frequencies of ultraviolet rays, gamma-rays and microwaves are ν_1 , ν_2 and ν_3 respectively. Then :
- (A) $\nu_1 = \nu_2 = \nu_3$ (B) $\nu_1 > \nu_2 > \nu_3$
 (C) $\nu_2 > \nu_1 > \nu_3$ (D) $\nu_3 > \nu_2 > \nu_1$
8. In the photoelectric effect, electrons are emitted from a given metal surface when the incident radiation has a certain minimum value of :
- (A) frequency (B) intensity
 (C) wavelength (D) angle of incidence
9. In Bohr model of a hydrogen atom, the increase in the angular momentum of the electron, when it is excited from $n = 1$ to $n = 3$ orbit, is :
- (A) $\frac{h}{2\pi}$ (B) $\frac{2h}{\pi}$
 (C) $\frac{3h}{2\pi}$ (D) $\frac{h}{\pi}$

10. A positively charged particle of mass m , moving with velocity v is bombarded on a heavy nucleus of charge Ze . Its distance of closest approach from the nucleus is proportional to :
- (A) $\frac{Z}{mv^2}$
- (B) $\frac{v^2}{Zm}$
- (C) v^2
- (D) $\frac{Zv^2}{m}$
11. The effect on width of depletion layer and height of the potential barrier of a p-n junction, when it is forward biased, are respectively :
- (A) decreases, decreases
- (B) increases, increases
- (C) decreases, increases
- (D) increases, decreases
12. The doping of a semiconductor with donor impurity results in :
- (A) formation of a p-type semiconductor
- (B) formation of a new energy level just above the filled valence band
- (C) decrease in the concentration of electrons
- (D) formation of a new energy level just below the conduction band

Questions number 13 to 16 are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false and Reason (R) is also false.

13. Assertion (A) : Work functions of different metals are different.

Reason (R) : Metals differ in their conductivity.

14. Assertion (A) : The capacitive reactance increases with the increase of the frequency (ω) of applied ac voltage.

Reason (R) : The value of capacitive reactance (X_C) is ωC .

15. Assertion (A) : When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

16. Assertion (A) : Signals through an optical fibre can be transmitted without any appreciable loss of energy.

Reason (R) : An optical fibre works on the principle of total internal reflection.

SECTION B

17. Two identical cells, each of emf E and internal resistance r , are connected in series with a resistor of $8\ \Omega$. A current of 1.5 A flows in the circuit. When these cells are connected in parallel, they send 1.0 A current through the same resistor. Calculate the internal resistance, r and emf, E of the cell. 2

18. (a) Name the factors on which focal length of a lens depends. How will a thin biconvex glass lens submerged in water behave ? Give reason for your answer. 2

OR

- (b) How will the fringe width be affected in a Young's double slit experiment with light of wavelength λ if :
- (i) the separation between the slits is reduced to half, keeping the distance of the screen from the slits constant ?
 - (ii) the distance of the screen from the slits is reduced to half, keeping the separation between the slits constant ? 2
19. The power of the combination of two thin convex lenses L_1 and L_2 , kept coaxially in contact, is $+10\text{ D}$. If the focal length of L_1 is four times that of L_2 , find their focal lengths. 2

- 20.** Differentiate between mass defect and binding energy per nucleon. The binding energy per nucleon is nearly constant in the range $30 < A < 170$. Why ? 2
- 21.** What are the important considerations for choosing dopants for silicon or germanium ? Justify your answer. 2

SECTION C

- 22.** (a) Define the terms – ‘temperature coefficient of resistance’ and ‘electrical conductivity’ of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. 3

OR

- (b) Define ‘drift velocity’. Show that resistivity of the material of a conductor is inversely proportional to the relaxation time for the free electrons in the conductor. 3
- 23.** Two parallel straight wires of lengths 15 m and 20 m carrying currents 6.0 A and 8.0 A respectively in opposite directions, are 10 cm apart. Calculate the force on a 10 cm long section of the second wire near its centre. Also mention the nature of the force between these wires. 3

- 24.** A bulb and an inductor of reactance X_L are connected in series with an ac source. How will the brightness of the bulb change when :
- (a) an iron rod is inserted in the inductor ?
 - (b) number of turns in the inductor is decreased ?
 - (c) a capacitor of capacitive reactance $X_C (= X_L)$ is connected in series with the inductor ?
- 3**
- 25.** What is meant by 'displacement current' ? Show that during charging of a capacitor, displacement current is equal to conduction current in the circuit.
- 3**
- 26.** Answer the following giving reasons :
- 3**
- (a) The speed of emitted electrons depends on the work function of the photosensitive surface, for the incident radiation of a given frequency.
 - (b) The stopping potential (V_0) increases linearly with the frequency of incident radiation, for a given photosensitive surface.
 - (c) The photoelectric current increases linearly with the intensity of incident radiation, for a given surface.
- 27.** Explain the origin of Lyman and Balmer series in the emission spectrum of hydrogen atom. How are the maximum and the minimum wavelength of lines in a series determined ?
- 3**
- 28.** Explain the formation of energy bands in solids and hence define conduction band and valence band.
- 3**

SECTION D

Questions number 29 and 30 are case study based questions. Read the following paragraphs and answer the questions that follow.

- 29.** In a homogeneous medium, light travels along a straight line path. But when a ray of light travels from one medium to another, it deviates from its initial path. This bending of light is called refraction of light. The extent to which a ray deviates in medium 2 from its initial path in medium 1, is determined by the refractive index of medium 2 with respect to medium 1. The refractive index is also related to the ratio of speeds of light in the two media.

The phenomenon of total internal reflection occurs when light passes from a denser medium into a rarer medium and the angle of incidence exceeds the critical angle for the given pair of media.

- (i) A ray of monochromatic light of wavelength λ and speed v travels from air into a glass slab. Inside the glass slab, 1
- (A) both λ and v decrease
(B) both λ and v increase
(C) λ decreases and v increases
(D) λ increases and v decreases
- (ii) (a) A ray of light travelling from air into a medium, is incident on the interface at an angle of 45° . If the ray deviates by 15° from its initial path, the refractive index of the medium is : 1
- (A) 2 (B) $\sqrt{2}$
(C) $2\sqrt{2}$ (D) $\sqrt{3}$

OR

- (ii) (b) A point source of light is kept at the bottom of a tank filled with a transparent liquid up to a height H . If the radius of the circular patch on the surface from which light can come out is $\frac{H}{\sqrt{3}}$, the refractive index of the liquid is close to : 1
- (A) 1.3 (B) 1.5
(C) 1.7 (D) 2.0
- (iii) In an optical fiber, the refractive indices of the core and cladding are μ_1 and μ_2 respectively. Then : 1
- (A) $\mu_1 = \mu_2$
(B) $\mu_1 < \mu_2$
(C) μ_1 is slightly greater than μ_2
(D) μ_1 is large compared to μ_2
- (iv) Light of wavelength λ travelling with speed c in air, enters a medium of refractive index μ . The frequency of light in the medium is : 1
- (A) $\frac{c}{\mu\lambda}$
(B) $\frac{\mu c}{\lambda}$
(C) $\frac{\mu\lambda}{c}$
(D) $\frac{c}{\lambda}$

- 30.** An electric dipole is a pair of equal and opposite point charges $+q$ and $-q$ separated by a distance $2a$. A dipole can be characterised by dipole moment determined by its charge and separation between the two charges. It is a vector quantity. By convention the direction of a dipole is taken to be the direction from $-q$ to $+q$. Like fields due to a point charge, fields due to a given dipole at a point can be found.

When a dipole is placed in a uniform external electric field, it experiences a torque. This torque tends to rotate the dipole and work is done in this process. This work is stored as the potential energy of the system.

- (i) An electric dipole is kept in a uniform electric field such that the dipole moment is not along the electric field. It will experience :

1

- (A) A net force and a torque
- (B) A net force but no torque
- (C) A torque but no net force
- (D) neither a net force nor a torque

- (ii) An electric dipole consists of charges $+q$ and $-q$, separated by distance $2a$. The force acting on a point charge placed at a distance x ($\gg a$) on the axis of the dipole is F . If the distance of the charge is doubled, the force will become :

1

- | | |
|--------------------|-------------------|
| (A) $\frac{F}{16}$ | (B) $\frac{F}{8}$ |
| (C) $\frac{F}{4}$ | (D) $\frac{F}{2}$ |

- (iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be : 1

- (A) p (B) $2p$
(C) $p\sqrt{2}$ (D) $\frac{p}{\sqrt{2}}$

- (iv) (a) A molecule with dipole moment 3.0×10^{-30} C-m is placed in a uniform electric field of 2×10^4 NC⁻¹. The maximum torque experienced by the molecule is : 1

- (A) 1.5×10^{-34} N-m
(B) 6.0×10^{-26} N-m
(C) 4.5×10^{34} N-m
(D) 6.0×10^{-34} N-m

OR

- (iv) (b) A electric dipole consists of two point charges $+2$ pC and -2 pC, separated by a distance 1 mm. It is kept in a uniform electric field of 2×10^5 NC⁻¹. The amount of work done in rotating it from its position of stable to unstable equilibrium will be : 1

- (A) 2×10^{-10} J
(B) 4×10^{-10} J
(C) 8×10^{-10} J
(D) 1.6×10^{-11} J

SECTION E

- 31.** (a) (i) A capacitor of capacitance C is charged to a potential difference V by a battery. The battery is disconnected and the separation between the plates is halved. How will the following quantities be affected ?

- (1) Capacitance of the capacitor
- (2) Electric field between the plates
- (3) Energy stored in the capacitor

Justify your answer in each case.

- (ii) A capacitor of $20\ \mu\text{F}$ is charged to $30\ \text{V}$. It is then connected to an uncharged $30\ \mu\text{F}$ capacitor. Calculate the charges on each capacitor in equilibrium.

5

OR

- (b) (i) Obtain an expression for electric potential at a distance r from a point charge Q .
- (ii) Two point charges of $10\ \mu\text{C}$ and $-5\ \mu\text{C}$ are placed at $(-3\ \text{cm}, 0)$ and $(6\ \text{cm}, 0)$ in an external electric field $E = \frac{A}{r^2}$, where $A = 1.8 \times 10^5\ \text{N C}^{-1}\ \text{m}^2$. Calculate the electrostatic energy of the system.

5

- 32.** (a) (i) Define the term 'self-inductance' of a coil. Obtain an expression for self-inductance of a solenoid of area of cross-section A , length L and having n turns per unit length.
- (ii) The current flowing in a coil of inductance 50 mH is reduced from 10 A to 0 in 20 ms . Calculate the change in flux linked with the coil. 5

OR

- (b) (i) Briefly explain the working of an ac generator. Obtain the expression for the instantaneous value of the emf induced in the generator.
- (ii) Calculate the angular frequency at resonance of a series LCR circuit with $R = 12 \Omega$, $L = 2.0 \text{ H}$ and $C = 32 \mu\text{F}$, connected to an ac source. 5

- 33.** (a) (i) What are coherent sources of light ? Briefly explain how a pair of coherent sources are obtained in a Young's double-slit experiment. Mention the conditions for constructive and destructive interferences in this experiment.
- (ii) An object is kept 20 cm from a convex mirror of radius of curvature 60 cm . Find the position of the image formed. Will the image be real or virtual ? 5

OR

(b) (i) Answer the following giving reasons :

- (1) In modern microscopes, multicomponent lenses are used for both the objective and the eyepiece.
- (2) In a compound microscope, both the objective and the eyepiece have small focal lengths.
- (3) When viewing through a compound microscope, our eyes should be positioned not on the eyepiece, but a short distance away from it for best viewing.

(ii) Two students are separated by a 6 m partition wall in a room 9 m high. If both light and sound waves can bend around obstacles, how is it that the students are unable to see each other, though they can talk easily ?

5